

Data Cleaning Workflow for the Reconciled Formula 1 Database

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Introduction

This document describes the methodological workflow adopted for the Data Cleaning phase of the Formula 1 Data Warehouse project.

The Data Cleaning phase is performed after the construction and population of the reconciled database and after the execution of the Data Quality Assessment.

The goal of this document is not to present the final cleaning results yet, but to define how the cleaning process will be structured, documented, and connected to the later Data Warehouse design phase.

In this project, the Data Quality Assessment identifies possible problems in the data, while the Data Cleaning phase decides how to handle them through explicit and documented rules.

Position of Data Cleaning in the Project Workflow

The general project workflow is structured as follows:

1. Raw data extraction from FastF1 and Ergast/Jolpica.
2. Re-engineering of raw data into a reconciled relational schema.
3. Population of the reconciled PostgreSQL database.
4. Data Quality Assessment on the reconciled database.
5. Data Cleaning based on explicit cleaning rules.
6. Conceptual design of the Data Warehouse.
7. Attribute tree construction and editing.
8. DFM Fact Schema design.
9. Star Schema implementation.
10. ETL from the cleaned reconciled database to the Data Warehouse.

The Data Cleaning phase therefore acts as a bridge between the reconciled database and the Data Warehouse implementation.

Difference Between Data Quality Assessment and Data Cleaning

Data Quality Assessment

The Data Quality Assessment does not modify the database.

Its goal is to inspect the data and detect possible issues according to a set of predefined rules.

Examples of checked dimensions are:

- completeness;
- uniqueness;
- validity;
- consistency;
- accuracy and plausibility;
- timeliness;
- referential integrity.

The output of the Data Quality Assessment consists of scorecards and issue files.
The most important output for the cleaning phase is:

`issues_all_tables.csv`

This file summarizes the detected issues for the different tables of the reconciled database.

Data Cleaning

The Data Cleaning phase uses the DQA outputs as evidence, but it does not blindly execute all the detected issues as automatic corrections.

Instead, each problem must be interpreted and connected to a documented cleaning rule.

The cleaning phase answers the following question:

Given a detected data quality issue, what should be done with the affected data?

Possible actions include:

- correcting a value;
- standardizing a value;
- converting a value to a consistent format;
- keeping the value unchanged but documenting the issue;
- adding a flag;
- excluding the row from the cleaned analytical dataset;
- moving the row to a rejected rows table;
- correcting an upstream re-engineering script and regenerating the data.

Difference Between Cleaning and Analytical Filtering

A fundamental distinction must be maintained between Data Cleaning and Analytical Filtering.

Data Cleaning

Data Cleaning handles data that are incomplete, inconsistent, invalid, duplicated, unreliable, or not suitable for further processing.

Examples:

- missing identifiers;
- invalid lap times;
- impossible negative values;
- duplicated records;
- inconsistent foreign keys;
- inaccurate laps;
- deleted laps.

Analytical Filtering

Analytical Filtering does not fix wrong data.

It selects a subset of valid data that is useful for a specific analysis or storytelling.

Examples:

- keeping only Ferrari, Red Bull, and Mercedes;
- keeping only dry conditions;
- excluding wet sessions from a specific performance comparison;
- excluding in-laps and out-laps when studying pure lap performance;
- selecting only specific circuits.

A wet lap is not necessarily a wrong datum. It may be a valid datum that is outside the analytical focus of the final storytelling.

Therefore, it should not be confused with a data quality error.

Input of the Data Cleaning Script

The Data Cleaning script will use two main categories of input.

Database Input

The script reads the tables from the pre-cleaning reconciled schema.

Possible schema names are:

- `public`;
- `reconciled_raw`;
- `reconciled_pre_cleaning`.

In this project, the preferred solution is to preserve the original reconciled schema and write the cleaned output into a separate schema.

Data Quality Assessment Output

The script also reads the files produced by the Data Quality Assessment.

The most important files are:

- `issues_all_tables.csv`;
- `dq_scorecard_by_table.csv`;
- possible detailed issue files by table or by dimension.

The issue file is used to identify the detected problems.

The scorecard file is mainly used for documentation, prioritization, and before/after comparison.

Output of the Data Cleaning Script

The Data Cleaning script should produce both cleaned database tables and audit outputs.

Cleaned Database Schema

The preferred output is a new PostgreSQL schema:

`reconciled_clean`

This schema contains the cleaned version of the reconciled tables.

For example:

- `reconciled_clean.season`;
- `reconciled_clean.grand_prix`;
- `reconciled_clean.session`;
- `reconciled_clean.driver`;
- `reconciled_clean.team`;
- `reconciled_clean.result`;
- `reconciled_clean.lap`;
- `reconciled_clean.weather`;
- `reconciled_clean.track_status`.

Cleaning Metadata

In addition to the cleaned tables, the script should generate metadata that explain what happened during cleaning.

Possible metadata outputs are:

- `cleaning_actions.csv`;
- `cleaning_summary_by_table.csv`;
- `cleaning_summary_by_rule.csv`;
- `rejected_rows.csv`;
- `before_after_scorecard.csv`.

These outputs are important because the cleaning process must be traceable and reproducible.

Why a Separate Post-Cleaning Schema Is Preferred

The original reconciled database should not be overwritten directly.

The preferred professional choice is to maintain two versions of the reconciled data:

- a pre-cleaning schema;
- a post-cleaning schema.

This allows the project to preserve the original data produced by the re-engineering phase and to compare them with the cleaned version.

`reconciled_pre_cleaning → reconciled_clean`

This design supports:

- reproducibility;
- auditability;
- comparison before and after cleaning;
- easier debugging;
- safer ETL development.

Types of Problems Handled by Data Cleaning

The problems detected by the Data Quality Assessment can belong to different categories.

Not all categories require the same treatment.

Re-engineering Errors

Some issues may be caused by errors in the extraction or re-engineering scripts.

Examples:

- wrong joins;
- missing identifiers caused by an incorrect merge;
- wrong primary key definition;
- wrong foreign key mapping;
- incorrect DQA rule.

These problems should not be hidden inside the cleaning script.

The correct solution is:

1. correct the upstream script;
2. regenerate the reconciled data;
3. reload the database;
4. rerun the Data Quality Assessment.

Automatically Correctable Errors

Some issues can be corrected automatically because the correction rule is deterministic and safe.

Examples:

- trimming spaces in textual columns;
- converting empty strings to `NULL`;
- standardizing boolean values;
- standardizing categorical values;
- converting time values to a consistent numerical unit;
- removing exact duplicated rows.

Non-Reliable Rows

Some rows may be formally present but not reliable for analysis.

Examples in the `lap` table:

- deleted laps;
- inaccurate laps;
- laps with missing lap time;
- laps with impossible sector times;
- laps with inconsistent lap time and sector sum.

These rows should not necessarily be deleted from the original schema.

In the cleaned schema, they can be:

- removed from the cleaned analytical table;
- kept with a usability flag;
- moved to a rejected rows output;
- documented in the cleaning log.

Missing Values Not Safely Recoverable

Some missing values cannot be corrected without inventing information.

Examples:

- missing weather observations;
- missing sector times;
- missing driver descriptive fields;
- missing result gaps.

In these cases, the cleaning script should not invent values unless a clear and documented rule exists.

Possible actions are:

- keep the value as `NULL`;
- add a missing-value flag;
- exclude the row only if the missing value is essential for the intended analysis;
- document the issue for manual review.

Valid Data Outside the Analytical Focus

Some data are valid but may not be useful for the final storytelling.

Examples:

- wet laps;
- laps under safety car;
- pit in-laps;
- out-laps;
- sessions outside the final analytical focus.

These are not necessarily cleaning errors.

They should usually be handled later through analytical filtering or through flags that allow the analyst to include or exclude them.

Cleaning Rule Structure

Each cleaning rule should be explicitly documented.

A possible structure is the following:

Rule ID	Table	Condition	Action
CR_COMMON_001	All tables	Textual value contains leading or trailing spaces	Apply trim operation
CR_COMMON_002	All tables	Empty string in nullable column	Convert empty string to NULL
CR_LAP_001	Lap	Deleted = true	Mark lap as not usable for performance analysis
CR_LAP_002	Lap	IsAccurate = false	Mark lap as not usable for performance analysis
CR_LAP_003	Lap	LapTime_ms is missing	Mark lap as incomplete or reject if lap time is required
CR_WEATHER_001	Weather	WindSpeed < 0	Mark value as invalid and review
CR_TRACK_001	Track Status	Missing status code	Mark row as incomplete and review

Cleaning Actions

For each applied rule, the script should record an action.

A possible action log structure is:

Rule ID	Table	Row Identifier	Column	Old Value	New Value Action
CR_LAP_001	lap	[placeholder]	deleted	true	is_usable_for_per = false

Rule ID	Table	Row Identifier	Column	Old Value	New Value Action
CR_LAP_002	lap	[placeholder]	is_accurate	false	is_usable_for_per = false
CR_COMMON_001	driver	[placeholder]	first_name	“ Lewis ”	“Lewis”

Physical Modification vs Flagging

Not every issue should be handled by physically deleting or overwriting data.

In some cases, it is better to add flags that describe the usability or quality status of a row.

Physical Modification

Physical modification is appropriate when the correction is safe, deterministic, and does not change the semantic meaning of the data.

Examples:

- trimming spaces;
- standardizing case;
- converting empty strings to NULL;
- converting boolean representations;
- removing exact duplicate rows.

Flagging

Flagging is appropriate when the row contains information that may still be useful, but should not always be used in analytical calculations.

Examples:

- deleted lap;
- inaccurate lap;
- pit lap;
- incomplete lap;
- suspicious lap;
- wet-condition lap;
- safety-car lap.

Possible flags for the lap table are:

- `is_deleted_lap`;
- `is_inaccurate_lap`;
- `is_pit_lap`;
- `is_complete_lap`;
- `is_clean_lap`;

- `is_usable_for_performance_analysis`;
- `cleaning_status`;
- `cleaning_reason`.

Reason for Using Flags

Using flags preserves information while allowing controlled analytical selection.

Instead of permanently deleting all problematic rows, the cleaned schema can keep the row and explicitly mark whether it should be used for a specific analytical purpose.

This makes the process more transparent and reversible.

Relationship Between Cleaning Flags and the Attribute Tree

The cleaned reconciled schema may contain additional attributes introduced during Data Cleaning.

These attributes can influence the later conceptual design.

However, not every cleaning flag should automatically become part of the final Fact Schema.

General Rule

The general rule is:

A cleaning flag should be included in the attribute tree only if it has analytical meaning or if it is useful to define the valid analytical grain of the fact.

Technical Flags

Technical flags are used only for audit and debugging.

Examples:

- `cleaning_rule_id`;
- `cleaning_timestamp`;
- `source_issue_id`;
- `cleaning_batch_id`.

These attributes should not usually be included in the DFM Fact Schema.

They belong to cleaning metadata.

Analytical Flags

Analytical flags describe the analytical usability or semantic condition of a row.

Examples:

- `is_clean_lap`;
- `is_usable_for_performance_analysis`;
- `is_pit_lap`;
- `is_inaccurate_lap`;
- `is_wet_condition`;
- `is_safety_car_lap`.

These attributes may be included in the attribute tree because they can support filtering, slicing, or interpretation.

How to Represent These Flags in the DFM

In the DFM, these flags should usually not be modeled as full hierarchical dimensions.

They can be represented as:

- descriptive attributes of the fact;
- degenerate dimensional attributes;
- a small context dimension, if several related flags are grouped together.

For example, for the *Lap Performance* fact, the following attributes may be attached directly to the fact:

- `is_clean_lap`;
- `is_usable_for_performance_analysis`;
- `cleaning_status`.

Alternatively, they can be grouped into a small dimension:

Lap Quality Context

This dimension may contain:

- `cleaning_status`;
- `is_clean_lap`;
- `is_pit_lap`;
- `is_inaccurate_lap`;
- `is_usable_for_performance_analysis`.

Project Decision Placeholder

In this project, the current decision is:

[Placeholder: decide whether lap quality flags are kept as descriptive attributes of the *Lap Performance* fact or grouped into a separate *Lap Quality Context* dimension.]

Suggested Cleaning Strategy by Table

Common Rules for All Tables

Common cleaning rules are applied to all tables.

Examples:

- trim textual columns;
- convert empty strings to NULL;
- standardize boolean values;
- standardize date and timestamp formats;
- remove exact duplicated rows;
- preserve primary key and foreign key consistency.

Driver and Team Tables

The `driver` and `team` tables contain master data.

Possible cleaning operations:

- standardize textual fields;
- check missing identifiers;
- check duplicate identifiers;
- avoid deleting drivers or teams unless the identifier is invalid;
- preserve stable master data for joins.

Grand Prix and Session Tables

Possible cleaning operations:

- check missing event dates;
- check valid session types;
- check consistency between sessions and Grand Prix;
- preserve the relationship between session and Grand Prix;
- do not invent missing sessions.

Lap Table

The `lap` table is the most important table for the project because it is the source table for the *Lap Performance* fact.

Possible cleaning operations:

- identify deleted laps;
- identify inaccurate laps;
- identify missing lap times;
- identify impossible negative times;
- check consistency between sector times and lap time;
- standardize tyre compound values;
- identify pit laps;
- add usability flags for performance analysis.

Weather Table

Possible cleaning operations:

- check impossible temperature values;
- check negative wind speed;
- standardize rainfall as boolean;
- keep missing weather values as `NULL` if they cannot be safely recovered;
- avoid inventing meteorological values.

Track Status Table

Possible cleaning operations:

- check missing status values;
- standardize status codes;
- standardize message categories;
- preserve time alignment with the related session;
- create derived flags for safety car, virtual safety car, red flag, or yellow flag conditions if needed.

Rejected Rows

Some rows may be excluded from the cleaned analytical dataset.

However, excluded rows should be documented.

A possible rejected rows structure is:

Table	Row Identifier	Rule ID	Reason	Action
lap	[placeholder]	CR_LAP_001	Deleted lap	Excluded from clean performance subset
lap	[placeholder]	CR_LAP_002	Inaccurate lap	Excluded from clean performance subset
weather	[placeholder]	CR_WEATHER_001	Invalid wind speed	Marked for review

Validation After Cleaning

After the cleaning process, the Data Quality Assessment should be executed again on the cleaned schema.

This allows comparison between:

- pre-cleaning scorecards;
- post-cleaning scorecards;
- pre-cleaning issue counts;
- post-cleaning issue counts.

The goal is not necessarily to obtain a perfect score for every table.

The goal is to show that the remaining issues are either:

- expected;
- documented;
- not safely correctable;
- outside the scope of automatic cleaning.

Before and After Comparison

A summary table should be produced to show the effect of the cleaning phase.

Table	Issues Before	Issues After	Score Before	Score After
season	[placeholder]	[placeholder]	[placeholder]	[placeholder]
driver	[placeholder]	[placeholder]	[placeholder]	[placeholder]
team	[placeholder]	[placeholder]	[placeholder]	[placeholder]
lap	[placeholder]	[placeholder]	[placeholder]	[placeholder]
weather	[placeholder]	[placeholder]	[placeholder]	[placeholder]
track_status	[placeholder]	[placeholder]	[placeholder]	[placeholder]

Connection with the ETL Toward the Data Warehouse

The cleaned reconciled schema becomes the input of the ETL process toward the Data Warehouse.

The ETL should not be based on the pre-cleaning schema.

The preferred flow is:

`reconciled_pre_cleaning` → Data Quality Assessment → `reconciled_clean` → ETL →
`data_warehouse`

During the ETL, the project can decide whether to:

- load only clean and usable laps;
- load all laps but preserve quality flags;
- create a specific analytical subset;
- create dimensions and fact tables based on the final DFM and Star Schema.

Open Design Decisions

The following decisions must be completed during the implementation phase:

1. Which schema name should be used for the pre-cleaning data?
2. Which schema name should be used for the cleaned data?
3. Which cleaning rules should physically modify values?
4. Which cleaning rules should only add flags?
5. Which rows should be rejected?
6. Which flags should remain only in cleaning metadata?
7. Which flags should enter the attribute tree?
8. Which flags should be included in the DFM as descriptive attributes or context dimensions?
9. Should the ETL load only clean laps or all laps with quality flags?

Conclusion

The Data Cleaning phase is designed as a controlled and documented transformation process.

The Data Quality Assessment identifies possible problems, while the Data Cleaning phase applies explicit rules to correct, standardize, flag, reject, or document the affected data.

The original reconciled database should be preserved, and the cleaned data should be written into a separate schema.

This strategy supports traceability, reproducibility, and a safer transition toward the Data Warehouse.

The most important modeling decision is that cleaning flags become part of the cleaned reconciled schema, but they are included in the DFM only when they have analytical value or support the definition of the valid analytical grain of the fact.